Solutions to the Midterm

October 2024

1. Company A spent an additional \$5 million on sales and marketing expenses, increased its revenue by \$3 million and decreased its depreciation expense by \$2 million, thus reducing capital expenditures by \$2 million. Company A's pre-tax income is subject to a 20% tax rate. By how much will its net income change? By how much will its cash balance change?

   A. Net income increases by 8 million, cash balance decreases by 2 million.

   B. Net income decreases by 8 million, cash balance increases by 2 million.

   C. No change in the net income, no change in the cash balance.

   D. **No change in the net income, cash balance decreases by 2 million.**

   Solution: D. In the Income statement, the Revenue line increases by 3. The SG&A line increases by 5, and the Depreciation expense line decreases by 2. Therefore, the EBIT does not change (because 3-5+2=0); neither will the net income.
   By definition, the Cash Flow from Operations equals the net income plus the Depreciation expense, minus Accounts Receivable plus Accounts Payable. The Net Income has not changed; the Depreciation Expense has decreased by 2; Accounts Receivable and Payable haven't changed. Thus, the Cash from Operation will decrease by 2. CAPEX decreased by 2 million dollars; thus, the Cash from investing activities increases by 2 million. It follows that the Cash balance changes by $-2+2=0$.

2. Company B seeks to acquire some technology. It can do so via two means. Option 1: Develop the technology in-house; the estimated R&D cost is 5 million dollars. Option 2: Acquire at the price of 7 million dollars start-up C which already developed the technology; Company B will issue new shares to raise the 7 million. Currently, Company B has 10 million shares outstanding and a share price of \$10. It is subject to a 20% tax rate on pre-tax income. At present, its EPS is \$1. What is the EPS in each case?

   A. **With Option 1, EPS=0.6; With Option 2, EPS=0.93.**

B. With Option 1, EPS=0.5; With Option 2, EPS=0.93.

C. With Option 1, EPS=0.6; With Option 2, EPS=0.59.

D. None of the above.

Solution: A. Option 1 decreases the EBIT by 5, thus decreases the net income by $5 \times 0.8 = 4$. The number of shares remain constant at 10. The current EPS being 1, we conclude that the current net income is 10. Thus the net income with Option 1 will be 6. Thus, the EPS will be 6/10=0.6. Option 2 does not change the net income (will still be 10), however, the number of shares with increase by $\frac{7M}{10} = 0.7$. The new number of shares will be then 10.7. Thus, the EPS will be 10/10.7=0.93.

3. In 2023, Company D had an operating margin of 20% and Revenues of 10 million dollars. On the 1st of January 2021, Company D contracted a 10 million dollar loan subject to a 10% interest rate. On the 31st of December of each year from 2021, Company D repays 10% of the remaining balance on the loan. The pre-tax income is taxed at 50%. What is Company D's net income in 2023?

   A. 0 dollar.

   B. 5 million dollars.

   C. 1.19 million dollars.

   D. **0.595 million dollars.**

   Solution: D. The Gross Profit is 2. As we are not given any other information, we can set the EBIT to 2. To get the pre-tax income, we have to deduct interest payment from the EBIT. In 2023, the balance on the loan is 8.1. Thus, the interest payment in 2023 is $0.1 \times 8.1 = 0.81$. Thus, the pre-tax income is $2 - 0.81 = 1.19$. The net income is then $0.5 \times 1.19 = 0.595$.

4. Consider the following two securities. Both pay off 100 dollars when the economy is strong (with a probability of 0.8) and 0 when the economy is weak (with a probability of 0.2). Security A retails for a price of 50 dollars. Security B has a risk premium of 40%. The risk-free rate is 10%. Which arbitrage opportunity is available, and what is its associated profit?

   A. Short-sell A, buy B. Arbitrage profit of 3.30 dollars.

   B. **Short-sell B, buy A. Arbitrage profit of 3.30 dollars.**

   C. Short-sell A, buy B. Arbitrage profit of 7.14 dollars.

   D. Short-sell B, buy A. Arbitrage profit of 7.14 dollars.

   Solution: B. We conclude from the data that the return on B is 50%. This means that the price of B solves:

$$0.5 = \frac{80}{p} - 1.$$

Re-arranging, you find that $p = \frac{160}{3}$. The arbitrage profit will come from selling B and buying A (and 160/30-50=3.3).

5. Suppose you invested 50 dollars of your money in Treasury bills and an additional 50 dollars in Security A. Security A pays off 100 dollars when the economy is strong (with a probability of 0.8) and 0 when the economy is weak (with a probability of 0.2). What is the standard deviation of your portfolio?

   A. 0

   B. 40

   C. **20**

   D. 28.28

   Solution: C. The volatility of your portfolio equals x, the proportion of your money invested in Security A, times the volatility of security A. In this question, $x = \frac{1}{2}$. The volatility of A is:

$$SD(R_A) = \sqrt{0.8(100-80)^2 + 0.2(0-80)^2} = 40.$$

   Thus, $xSD(R_A) = 20$.

6. You currently have 100 dollars invested in some portfolio whose expected return is 60% and whose volatility is 40%. The risk-free rate is 5%, and the tangent portfolio has an expected return of 50% and a volatility of 30%. If your objective is to maximize the expected return from your portfolio, holding its volatility constant at 40%, which portfolio should you get instead of your current one, and what would its return be?

   A. **Borrow 33.33 dollars and buy for 133.33 dollars' worth of the tangent portfolio. The return from this portfolio is 65%.**

   B. Hold 33.33 dollars' worth of Treasury bills and 66.67 dollars' worth of the tangent portfolio. The return from this portfolio is 65%.

   C. Hold 100 dollars' worth of the tangent portfolio. The return from this portfolio is 50%.

   D. None of the above.

   Solution: A. Let's first determine the equation of the market line, which give you returns as a function of volatility. This is:

$$R = 0.05 + \frac{0.5-0.05}{0.3} \times SD(R)$$

   Thus, if you want to keep the volatility at 40%, the highest return you can earn is:

$$0.05 + \frac{0.5-0.05}{0.3} \times 0.4 = 0.65.$$

Let's determine the portfolio $P = (xTP, (1-x)TB)$, where TP denotes the tangent portoflio, TB denotes Treasury Bills, and x is the proportion of your 100 dollars you need to invest in TP. We want that:

$$SD(P) = 40\%.$$

We know that an expression for $SD(P)$ is: $SD(P) = xSD(TP) = 0.3x$. Thus, $x = \frac{0.4}{0.3} = 1.333$. If you invest $4/3$ of your 100 dollars in P, it means that you borrowed $\frac{100}{3}$ dollars.

7. You currently own a portfolio with 50% of your money placed in risk-free Treasury bills paying a 5% return and the remaining 50% in the Alpha fund. The expected return of the Alpha fund is 30%, and its volatility is 10%. You identify a new fund, the Gamma fund, whose expected return is 10% and whose volatility is 40%. For which values of the correlation between the returns of the two funds does adding the Gamma fund improve your portfolio?

   A. There are no such values.

   B. Above 0.10.

   C. **Below 0.05.**

   D. Below 0.

   Solution: C. Investing in the Gamma fund improves the Share ratio of your portfolio if its return of 10% exceeds the minimum return that equals:

   $$0.05 + \beta_{Gamma}(0.3 - 0.05).$$

   We thus need to calculate the β of the Gamma fund. This is:

   $$\beta_{Gamma} = \frac{SD(R_{Gamma}) \times Corr(R_{Gamma}, R_{Omega})}{SD(R_{Omega})} = 4 \times Corr(R_{Gamma}, R_{Omega}).$$

   Hence, investing in the Gamma fund improves your Sharpe ratio if, and only if,

   $$0.1 \geq 0.05 + 4 \times Corr(R_{Gamma}, R_{Omega}) \times (0.3 - 0.05)$$

   Re-arranging, the above is equivalent to:

   $$Corr(R_{Gamma}, R_{Omega}) \leq 0.05.$$

8. Which statement is true? According to the CAPM, if the market risk premium is 10%,

   A. The beta of a stock whose return is the risk-free rate is strictly negative.

   B. **The beta of a stock whose excess return (compared to the risk-free rate) equals the market risk premium is 1.**

C. The beta of a stock whose excess return (compared to the risk-free rate) is strictly greater than the risk premium is strictly lower than 1.

D. None of the above statements are correct.

Solution: B. According to the CAPM, the return of a stock (denoted by R_s) is linked to the market risk premium (MRP) via the following relation:

$$R_s = r_f + \beta_s \times MRP,$$

where β is the marginal risk from the stock in the market portfolio. The excess return of stock s is $R_s - r_f$. Thus, the CAPM equation tells us that:

$$R_s - r_f = \beta_s \times MRP.$$

If $R_s - r_f = MRP$, it means that $\beta_s = 1$.

9. Company E has 30 million shares outstanding trading for 10 dollars per share. The company also has 200 million dollars in outstanding debt. Suppose Company E's equity cost of capital is 10%, its debt cost of capital is 5%, and the corporate tax rate is 20%. What is Company E's WACC?

   A. 6.8%.

   B. 5%.

   C. 10%.

   D. **7.6%.**

   Solution: D. By definition,

$$r_{wacc} = \frac{E}{E+D} r_E + \frac{D}{D+E} r_D (1 - \tau_C),$$

where E is the present value of Equity, D is the present value of the Debt, and τ_C is the corporate tax rate. Given the data, $E = 30 \times 10 = 300$ million dollars. Also, $D = 200$ million dollars. Hence, $E + D = 500$ million dollars. Also, $\tau_C = 0.2$, $r_E = 0.1$, $r_D = 0.05$. Plugging in these values, one gets:

$$r_{wacc} = \frac{300}{500} \times 0.1 + \frac{200}{500} \times 0.05 \times 0.8 = 0.076$$

10. Assume that Croatia, one of the European countries that uses the euro as its currency, would prefer that Euro would be depreciated against the dollar. Can it apply Croatian central bank intervention to achieve this objective?

   A. Yes.

   B. Maybe.

   C. **Only the ECB can influence the FX rate of the EURO.**

Solution: C.

11. Assume the following information:

	Bank 1	Bank 2
Bid price of New Zealand dollars	\$0.398	\$0.401
Ask price of New Zealand dollars	\$ 0.400	\$0.404

Given this information, is locational arbitrage possible? If so, explain the steps involved in locational arbitrage, and compute the profit from this arbitrage if you had \$1,000,000 to use. How much is the profit?

A. **\$2,500.**

B. \$-14,500.

C. \$15,075.

D. \$10,000.

Solution: A. One could purchase New Zealand dollars at Bank 1 for \$0.40 and sell them to Bank 2 for \$0.401. With \$1 million available, 2.5 million New Zealand dollars could be purchased at Bank 1. These New Zealand dollars could then be sold to Bank 2 for \$1,002,500, thereby generating a profit of \$2,500.

12. Assume the following information:

	Quoted price
Spot rate of Canadian dollar	\$0.80
90-day forward rate of Canadian dollar	\$0.79
90-day Canadian interest rate	4%
90-day U.S. interest rate	2.5%

What would be the nominal yield (percentage return) to a U.S. investor who used covered interest arbitrage? (Assume the investor invests \$1,000,000.)

A. 1.5%.

B. **2.7%.**

C. 0.2%.

Solution: B. $\$1,000,000/\$0.80 = C\$1,250,000 \times (1.04) = C\$1,300,000 \times \$0.79 = \$1,027,000$.
Nominal Yield $= (\$1,027,000 - \$1,000,000)/\$1,000,000 = 2.7\%$, which exceeds the yield in the U.S. over the 90-day period.
If we pay attention to the USD possibilities 2.7%-2,5%=0,2%.

13. Boston Co. will receive 1 million euros in one year from selling exports. It did not hedge this future transaction. Boston believes that the future value of the euro will be determined by purchasing power parity (PPP). It expects that inflation in countries using the euro will be 12% next year, while inflation in the U.S. will be 7% next year. Today the spot rate of the euro is $1.46, and the one-year forward rate is $1.50. Estimate the amount of U.S. dollars that Boston will receive in one year when converting its euro receivables into U.S. dollars.

    A. **$1,394,800.**

    B. $1,528,224.

    C. $1,460,000.

    D. $1,500,000

    Solution: A. Expected appreciation of euro $= (1+0.07)/(1+0.12)-1 = -0.04464$.
    The expected spot rate of euro in one year $= 1.46 \times (1-0.04464) = \$1.3948$.
    Expected dollars received from receivables $= 1,000,000 units \times \$1.3948 = \$1,394,800$.

14. Consider two firms, With and Without, that have identical assets that generate identical cash flows. Without is an all-equity firm, with 1 million shares outstanding that trade for a price of $24 per share. With has 2 million shares outstanding and $12 million dollars in debt at an interest rate of 5%, has no plans to change the capital structure According to MM Proposition 1, the stock price for With is closest to:

    A. $8.

    B. $24.

    C. **$6.**

    D. $12.

    Solution: C. Equity Value of With=Asset Value of With $= \$24 \times 1mln = \$24mln$.
    Equity Value of Without= $24-12 = \$12mln$.
    Share price of Without $= \$12mln/2mln = \$6$.

15. Which of the following is TRUE?

    A. In perfect capital markets and with no change in cash flows an increase in leverage will increase the value of equity.

    B. In perfect capital markets and with no change in cash flows an increase in leverage will increase the value of the company.

    C. In perfect capital markets and with no change in cash flows an increase in leverage will increase the value of equity if the debt is risk-free.

D. **In perfect capital markets and with no change in cash flows an increase in leverage will increase the cost of equity.**

Solution: D.

16. Fovam Corporation has a current share price of \$6 and 10 million shares outstanding. Fovam announces to borrow \$20 million and repurchase shares. Fovam pays a corporate tax rate of 40%, and shareholders expect the change in debt to be permanent. Suppose the only imperfections are corporate taxes and financial distress costs. If the share price rises to \$7.8 after this announcement, what is the PV of financial distress costs (in millions of dollars) Fovam will incur as the result of this new debt?
Solution:

$$0.4 \times \frac{20}{10} + 6 = \$6.8 \text{ per share.}$$

Now, $(7.8 - 6.8) \times 10 = \$1$ million.

17. Which of the following is False?

A. **In perfect capital markets increasing leverage is creating value to shareholders because of the higher earnings per share.**

B. In perfect capital markets, leverage increases the risk of equity.

C. In perfect capital markets the cost of equity increases with leverage.

D. In perfect capital markets, the total cost of capital does not change with leverage.

Solution: A.

18. PPP Corporation's management team is meeting to decide on a new corporate strategy. There are four options, each with a different probability of success and total firm value in the event of success, as shown below:

	A	B	C	D
Probability of Success	1	0.8	0.6	0.4
Firm Value if Success	50	60	70	80

Assume that for each strategy, firm value is zero in the event of failure. Suppose Petron Corp. has debt with a face value of \$35 million outstanding. For simplicity, assume all risk is idiosyncratic, the risk-free interest rate is zero, and there are no taxes. How much will equity holders gain or lose by recapitalising to reduce its debt to \$15 million? What type of agency problem does this example illustrate?

A. \$5 mln gain, cashing out.

B. **\$5 mln loss, leverage ratchet effect of debt overhang.**

C. \$8 mln loss, leverage ratchet effect of debt overhang.

D. \$8 mln gain, cashing out.

Solution: B.

	A	B	C	D
Probability of Success	1	0.8	0.6	0.4
Firm Value if Success	50	60	70	80
Debt	35	35	35	35
Equity payoff if successful	15	25	35	45
Equity value	15	20	21	18
Debt after reduction	15	15	15	15
Value of Equity from Project (payoff x prob of success)	35	36	33	26
Addition equity to pay back debt	-20	-20	-20	-20
	15	16	13	6

19. Corporation XYZ has an AAA rated bond with a yield to maturity equal to 12%. XYZ's beta is 0.55, Treasury bill rate is 5%, and the expected market rate of return is 15%. What is the default premium the XYZ's bond?

   A. **1.5%.**
   B. 10.5%.
   C. 10%.
   D. 5.5%.

   Solution: A.

$$E[r] = 5\% + 0.55 \times (15\% - 5\%) = 10.5$$

   Default premium=12-10.5=1.5.

20. Which of the following statements is FALSE?

   A. Given a forecast of future interest payments, we can determine the interest tax shield and compute its present value by discounting it at a rate that corresponds to its risk.
   B. **When there is excess cash it is used first, to pay out to shareholders and then to repay debt.**
   C. When there is a need for external financing equity is issued first and then debt.
   D. In the presence of information asymmetry issuing stock emits a positive signal to the market.

   Solution: B.

21. Which of the following is true according to the pecking order theory?

A. There is a trade-off between benefits and the costs of leverage.

B. **When there is excess cash it is used first, to pay out to shareholders and then to repay debt.**

C. When there is a need for external financing equity is issued first and then debt.

D. In the presence of information asymmetry issuing stock emits a positive signal to the market.

Solution: B.

22. Which of the following is FALSE?

A. **In perfect capital markets dividend payout is value-destroying because the ex-dividend price is lower than cum-dividend price.**

B. In perfect capital markets investors are indifferent between share-repurchasing and dividend payouts.

C. In perfect capital markets share price does not change after share repurchase because of the decrease in the number of shares and the value of total equity.

D. In perfect capital markets shareholders are indifferent between dividend-paying and not-paying firms.

Solution: A.

23. Wcorp has assets with a market value of $600 million, $70 million of which are cash. It has a debt of $250 million, and 20 million shares outstanding. Assume perfect capital markets. If Wcorp distributes the $50 million as a share repurchase, then its stock price and debt-to-equity ratio after the share repurchase will be closest to:

A. **Share price $17.50, debt/equity 0.833.**

B. Share price $17.14, debt/equity 0.833.

C. Share price $17.14, debt/equity 0.714.

D. Share price $17.50, debt/equity 0.714.

Solution: A. If Assets=Debt+Equity, then the value of Equity is: $600-250=350$. Since the value of Equity is the share price times the number of shares outstanding, the share price is $\frac{350}{20}=17.5\$$. When capital markets are perfect, a share repurchase does not change the share price; hence, after the operation, a share will still trade for 17.5$. The new debt-to-equity ratio will be:

$$\frac{250}{350-50}=0.8333.$$

24. If the only market imperfection is taxation and all shareholders are individual investors under the same tax regime (the same income tax bracket, no loss-making) shareholders prefer share repurchases when (A) the tax rate on dividends > the tax rate on capital gains, (B) When the tax rate on dividends = the tax rate on capital gains.

   A. A.
   B. B.
   C. **A and B.**
   D. Neither A nor B.

   Solution: C.

25. How do investors interpret share repurchase announcements?

   A. **Positive signal that the stock is underpriced.**
   B. Negative signal that the stock is overpriced.
   C. Negative signal that the stock is underpriced.
   D. Positive signal that debt is overpriced.

   Solution: A.